

ABSTRACT

Groundwater is the main source of water for drinking and irrigation in Song and environs, Adamawa State. Thirty groundwater samples were analysed to assess suitability for domestic and agricultural use. Physicochemical parameters were determined using standard methods and compared with WHO drinking water guidelines. Irrigation suitability was evaluated using sodium adsorption ratio (SAR), magnesium hazard (MH), Kelly's ratio (KR), percent sodium (%Na⁺), and permeability index (PI). Results showed pH ranged from 4.6–6.3, indicating moderately acidic water below the

H YDROGEOCHEMICAL AND PHYSICOCHEMICAL STUDIES OF GROUNDWATER IN SONG AND ENVIRONS, ADAMAWA STATE, NORTH- EASTERN NIGERIA

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Introduction

Nowadays, the major source of water for drinking, agricultural and industrial use is groundwater. In the last few decades, the use of groundwater has increased significantly due to its widespread occurrence and high standard. The contribution from groundwater is crucial in global water supply. Over forty percent of the world's food is produced by irrigated agriculture that relies largely on groundwater (Muhammad, et. al., 2019) and about two billion people depend directly upon aquifers for drinking water. In northeastern Nigeria, urban growth characterized by population influx and agricultural activities pose serious water shortages and concomitant health hazards on groundwater quality (Olalekan, et. al., 2020). This concern has attracted overwhelming attention of civil engineers and hydrogeologists; because public water supply is inaccessible or inadequate, unreliable and irregular. As a result, people resort to groundwater abstracted through hand dug wells and borehole systems (Muhammad, et. al., 2019).

Water quality is affected by its dissolved geochemical constituents. The concentrations of the dissolved constituents have to be correlated with the standards of the global organizations charged with the responsibility of providing safe



WHO limit of 6.5–8.5. Electrical conductivity, total dissolved solids, and total hardness ranged from 142–452 $\mu\text{S}/\text{cm}$, 110–301 mg/L, and 19.65–98.26 mg/L, respectively, and were within acceptable limits. Nitrate exceeded WHO limits in 5 wells, while iron exceeded limits in 27% of samples due to underlying volcanic rocks. SAR, KR, and %Na values indicated excellent suitability for irrigation, but 12 wells had $\text{MH} > 50$ and are not recommended for irrigation. The study concludes that groundwater is generally suitable for irrigation but requires treatment before drinking in some locations.

Keywords: Groundwater, water quality, hydrochemistry, physico-chemical analysis, drinking water, irrigation.

drinking water to people (Zhou, et. al., 2020). Water quality is highly affected by the geochemical processes occurring within the groundwater and reactions with aquifer minerals and rocks (Li, et. al., 2021). Groundwater origin and history of its passage through underground materials, with which water has been in contact with, also have effects on its hydrogeochemical composition. Dissolved minerals from the soil and rock layers through which groundwater passes are contained in it (Paternoster, et. al., 2021). In addition, some harmful contaminants through the process of seepage from the surface water and biological activities may also be present (El Osta, et. al., 2022). The mineral composition of the aquifer through which groundwater flows is an indicator to its chemistry. Geology and chemical characteristics of the aquifer are important parameters on which both hydrochemical processes and hydrogeochemistry depend; and these vary spatially and temporally (Olalekan, et. al., 2020). The chemical composition of groundwater is controlled by the Hydrogeochemical processes such as dissolution, precipitation, ion exchange processes and the residence time along the flow path (Paternoster, et. al., 2021). This research attempts to investigate the hydrogeochemical and physico-chemical parameters of groundwater in the study area and evaluate its quality and standards with the aim of assessing the geochemical processes affecting the groundwater chemistry.

Problem Statement

Song; the headquarters of Song Local Government Area of Adamawa State, is a fast growing town; where the main sources of water supply is groundwater is obtained from boreholes. The water have a particular taste; hence the need to carry out a physicochemical assessment to determine its quality. Attention was not previously given to proper evaluation of water quality for human consumption in this area. Therefore, it becomes necessary is to conduct hydrogeochemical and physico-chemical analysis of the water in order to know its standard.

Aim and Objectives

The aim of this research is to investigate the physicochemical parameters of groundwater in the study area in order to evaluate its quality and standards for drinking and irrigation purposes. The objectives are as follows:

1. To determine the physicochemical parameters of groundwater in the study area.

2. To determine the main factors controlling water quality in the study area.
3. To evaluate the water suitability for domestic and irrigation water supply

Location of the Study Area

The study area lies between longitudes $12^{\circ} 36' E$ and $12^{\circ} 40' E$ and latitudes $9^{\circ} 48' N$ and $9^{\circ} 52' N$ and falls in the northeastern Nigerian basement complex (Figure 1). It has hilly topography with structural ridges and inselbergs surrounding the alluvial plains. The area is characterized by tropical dry and wet seasons; the rainy season lasts from April to October, while the dry season from November to March; the mean annual rainfall ranges from 800 mm to 900 mm (Adebayo, et. al., 2020).

Temperatures are high between March to June, when the daily maximum temperatures reaching up to $42^{\circ} C$, and also in October to November, immediately after the rainy season. During the months of December and January, temperatures drop to $20^{\circ} C$ due to obstruction of sun rays by harmattan dusts, and also between July to September; due the influence of the humid maritime air mass. The relative humidity rises up to 90% in the months of August and September (Adebayo, et. al., 2020).



Figure 1 Map of Adamawa State, Nigeria showing the Study Area

Geology of the Study area

The study area is a basement complex terrain; with rocks which witnessed both pre-pan African and pan-African orogenies (Ekwok, et. al., 2021). The surface geology has been weathered to superficial deposits of gravels sands and clayey soils; with outcrops of the basement rocks still preserved in the hills. The underlying rocks consist of gneisses, older granites and tertiary volcanics (Figure 2).

The gneisses are coarse-grained high-grade metamorphic rocks, with light and dark minerals segregated into visible bands; made up of biotite gneiss and banded gneiss. The biotite gneisses are fine-grained with strong foliation caused by the parallel arrangement of alternating dark and light minerals. The banded gneisses show alternating light-coloured and dark bands and exhibit intricate folding of their bands (Ekwok, et. al., 2021). These rocks exhibit fractures, veins, fissure, faults, joints and fractured zones due to tectonism. The older granites are coarse-grained igneous rocks made up of feldspar, mica, quartz, biotites, and microcline; occurring as large circular masses within the older gneiss complexes. The volcanic rocks occur as interbedded lavas among the basement complex and as post metamorphic extrusives and are composed largely of calcic plagioclases and ferromagnesian minerals (Ekwok, et. al., 2021). The most prominent are basalts; which and are characterized by small grain sizes and approximately equal proportion of calcium-rich plagioclase, feldspar and calcium rich pyroxenes. They possess vesicles and columnar jointing and occur as low ridges along stream channels of rivers as pumices, bombs and columnar joints (Ekwok, et. al., 2021).



Figure 2 Geologic Map of Adamawa State Showing the Rock Types Found in the Study Area

In the study area, groundwater occurs in the unconsolidated weathered overburden and fractured basement aquifers; which are heterogeneous and consist of layers of fine to coarse grained sands, silts, sandy clays and clays.

Materials and Methods

Water Sampling

Reconnaissance survey of the study area was carried out for a better understanding of the spatial distribution of boreholes. Groundwater samples were collected during the month of May, 2025; at thirty (30) sampling points using a random sampling technique. Nine (9) chemical parameters were analyzed for Total Suspended Solids, Total Hardness,



Calcium, Magnesium, Chloride, Nitrate, Sulphate, Copper and Iron, while four (4) physical parameters were measured for pH, Electrical Conductivity and Total Dissolved Solids using standard methods for water samples examination after American Public Health Association APHA (2012).

One litre of water was collected from each borehole; which was pumped for about three to five minutes prior to sample collection. Water samples were collected in two labeled, well-drained plastic containers which were tightly corked; to minimize contamination that could alter the water constituents.

The first container was acidified with two drops of concentrated nitric acid (HNO_3) for cations determination. The purpose of acidification is to stop bacterial growth, inhibit oxidation reactions and lead to precipitation of cations. The second plastic container was used for anion determination. The water samples were kept in cool boxes to keep the temperature below 20°C and later taken to the laboratory for analysis. The coordinates and elevation readings at borehole points were recorded using portable GARMIN Etrex 76 Global Positioning System GPS.

Water Analysis

The in situ physical parameters such as Temperature, Colour, Taste, Conductivity, and Total Dissolved Solid (TDS) were measured using the Ec/pH Multi parameter instrument; as the chemistry of groundwater is sensitive to environmental changes (Alsuhaime, et. al., 2019). pH buffer 7, pH buffer 4/10 and distilled water were used to calibrate the pH meter at the first instance. 50ml of the water sample was poured into a beaker and the probe inserted into the water 10 minutes after the meter was switch on and the pH was recorded when the reading became stable. The conductivity solution was used to calibrate the conductivity meter at 25°C ; which was then switched on and inserted into the 50ml water sample. When the reading became stable, the electrical conductivity value was recorded. A standard solution of sulphuric acid (H_2SO_4) was used to determine alkalinity titrimetrically. The sample was titrated with the 0.1N H_2SO_4 using phenolphthalein and methyl orange indicators. 100ml water was measured into a conical flask and 3 drops of phenolphthalein indicator added. When the sample remained colourless, it indicated in all cases that phenolphthalein alkalinity was zero.

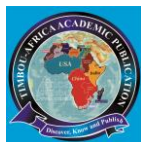
Standard methods prescribed by the American Public Health Association (2012) were used in analyzing the groundwater samples. Turbidimetric method was used to analyze sulphate. The method relies on the formation of Barium sulphate in the presence of acidified HCl and Barium chloride.

Assessing Drinking water quality

The two main yardsticks on which the standard for assessment of drinking water quality are the presence of substances with adverse health effects and the presence of objectionable taste, odours and colours (Khanoranga, 2019). The dissolved geochemical constituents in groundwater obtained in this study are correlated with the World Health Organization WHO (2021) standards.

Assessing irrigation water quality

The suitability of groundwater for irrigation was determined using the concentrations of sodium, calcium and magnesium (Yildiz, et. al., 2020). According to Singh, et. al., (2020), the



indicators for irrigation water quality include sodium adsorption ratio (SAR), magnesium hazard (MH), Kelly's ratio (KR), sodium percentage (Na %) and permeability index (PI); in addition to measured values of and electrical conductivity (EC), total dissolved solids and total hardness.

Sodium Adsorption Ratio (SAR)

Sodium Adsorption Ratio (SAR) gives the relative ratio of Na^+ , Ca^{2+} and Mg^{2+} ions contained in groundwater sample and is estimated using Equation (1). Rawat et al. (2018) noted that SAR influences the percolation time of water in the soil. Based on SAR, water is classified as excellent (< 10), good ($10 - 18$), doubtful ($18 - 20$) and unsuitable (> 26).

$$\text{SAR} = \text{Na}^+ / \sqrt{(\text{Ca}^{2+} + \text{Mg}^{2+})} \dots\dots\dots (1)$$

Magnesium Hazard (MH)

The soil quality is affected by excess concentration of magnesium in groundwater, since a high value of Mg^{2+} in water increases soil pH and leads to a decrease in the availability of phosphorus (Batarseh, et. al., (2021). MH is estimated using Equation (2).

$$\text{MH} = \text{Mg}^{2+} / (\text{Ca}^{2+} + \text{Mg}^{2+}) * 100 \dots\dots\dots (7)$$

Kelly's Ratio (KR)

KR is based on the concentration of Na^+ versus Mg^{2+} and Ca^{2+} ; is estimated using Equation (3).

$$\text{KR} = \text{Na}^+ / (\text{Mg}^{2+} + \text{Ca}^{2+}) \dots\dots\dots (3)$$

Percentage Sodium (%Na⁺)

%Na⁺ gives the percentage composition of Na⁺ in water; is estimated using Equation (4).

$$\% \text{Na}^+ = \text{Na}^+ / (\text{Na}^+ + \text{K}^+ + \text{Mg}^{2+} + \text{Ca}^{2+}) \dots\dots\dots (4)$$

Permeability Index (PI)

PI is the effect of irrigation water quality on soil permeability; is estimated using Equation (4).

$$\text{PI} = [(\text{Na}^+ + \sqrt{\text{HCO}_3^-}) / (\text{Ca}^{2+} + \text{Mg}^{2+} + \text{Na}^+)] * 100 \dots\dots\dots$$

Results and Discussion

Physical Parameters

Temperature, pH, TDS and conductivity are the important physical parameters of concern that were examined in this research and the results are shown in Table 1.

Table 1: Physical parameters of groundwater in the study area

Sample location	Temperature (°C)	pH	Electrical (μs/cm)	Conductivity	Total Solids	Dissolved
SO01	18.87	4.6	228		162	
SO02	20.21	6.0	269		177	
SO03	19.55	4.6	282		193	
SO04	18.77	5.7	284		128	
SO05	21.26	5.4	319		210	
SO06	20.71	4.8	350		243	
SO07	19.45	5.6	195		142	



Sample location	Temperature (°C)	pH	Electrical (µs/cm)	Conductivity	Total Solids	Dissolved
SO08	18.75	5.5	177		121	
SO09	18.24	6.2	165		112	
SO10	19.74	6.0	198		130	
SO11	19.42	4.6	207		145	
SO12	21.21	4.7	221		150	
SO13	20.64	5.2	410		301	
SO14	18.68	6.1	385		273	
SO15	19.42	6.3	424		285	
SO16	21.52	4.8	452		301	
SO17	20.28	5.8	394		287	
SO18	20.76	6.1	243		174	
SO19	18.87	6.2	142		167	
SO20	18.54	5.3	159		105	
SO21	20.21	6.0	172		128	
SO22	18.65	5.8	225		154	
SO23	19.44	4.9	247		163	
SO24	18.32	4.7	264		172	
SO25	21.12	5.4	253		170	
SO26	18.56	5.7	276		173	
SO27	21.22	5.3	305		211	
SO28	19.70	4.8	155		108	
SO29	20.32	5.3	162		110	
SO30	18.75	5.9	291		186	

The pH value ranges from 4.6 in wells SO01, SO03 and SO11 to 6.3 in well SO15 with average value of 5.43. These revealed that the groundwaters are moderately acidic; the low pH values could be influenced by the high level of anthropogenic activities. When surface water intermingled with groundwater, the contaminants in the surface water are introduced into the groundwater system (Tóth, 2020; Zhou *et al.*, 2020). It is recommended that the value of pH in potable water should not exceed the range of 6.5 – 8.5; which implies that groundwater in the study area in terms of pH value is not suitable for drinking purposes because all the groundwater samples recorded pH values less than 6.5 (WHO, 2021). The values of temperature vary between 18.24 °C in well SO09 and 21.52 °C in well SO16 with an average of 19.71°C; which are all below the recommended values of 30°C to 35°C. The values of Total Dissolved Solids (TDS) range from 110 in well SO29 to 301 in wells SO13 and SO16 with an average value of 153.63; indicating that the groundwater in the study area is fresh water. The TDS values are all within the recommended limits and hence are within the non-saline class. According to Yetiş *et al.* (2019) and Akakuru *et al.* (2021) variation of TDS is greatly controlled by the anthropogenic activities, geochemical processes taking place within the groundwater repositories and the residence time of groundwater in a hydrogeologic unit (Alsuhaime *et al.* 2019). The electrical conductivity (EC) values of the water samples vary from 142 µs/cm in well SO19 to 452 µs/cm in well SO16 with average value of 249.95 µs/cm. These values are low and



fall within the recommended standards; excellent to good quality water for drinking purpose (Akakuru et al. 2021).

Chemical Parameters

The anions analyzed include; bicarbonate ion (HCO_3^-), chloride ion (Cl^-), nitrate ion (NO_3^-) and sulphate ion (SO_4^{2-}). The chloride ion (Cl^-) values range from as low as 8.65 mg/L in well SO17 to as high as 96.32 mg/L in well SO18 with average value of 28.01 mg/L; which implies that the Cl^- the values were within the WHO (2021) permissible limits. Excess Cl^- will produce salty taste, which is not good for human consumption (Rahman et al. 2015). Cl^- concentration in groundwater may be due to the presence of chlorides from rocks, or contamination by domestic sewage (Saha, et al. 2019). The low concentration of Cl^- is attributed to low salinity of the groundwater of the study area. Bicarbonate ion (HCO_3^-) concentration ranges from 142 mg/L in well SO22 to 251 mg/L in well SO18 with an average value of 185.87 mg/L while the nitrite ions (NO_3^-) values ranges from 6.99 mg/L in well SO17 to 75.15 mg/L in well SO19 with a mean of 30.47 mg/L. The concentrations of NO_3^- are above WHO limits in wells SO07, SO15, SO16, SO18 and SO19. The relatively high NO_3^- content in these wells could be related to a longer residence time of the groundwater. Water from these wells is therefore characterized by high risk of nitrate contamination. High nitrate concentration is responsible for the blue baby syndrome (methemoglobinemia) (Karakus and Yildiz, 2020).

Table 2: Chemical parameters of groundwater in the study area

Sample location	HCO_3^- (mg/l)	CO_3^{2-} (mg/l)	TH (mg/l)	Fe^{2+} (mg/l)	F (mg/l)	Cl ⁻ (mg/l)	NO_2^- (mg/l)	Mg^{2+} (mg/l)	Na^+ (mg/l)	Ca^+ (mg/l)	SO_4^{2-} (mg/l)	K ⁺ (mg/l)	Cu^{2+} (mg/l)
SO01	173	3.02	46.37	0.012	0.38	9.53	7.65	25.84	1.48	20.39	21.92	2.72	0.05
SO02	193	3.10	50.42	0.016	0.41	10.88	8.19	27.21	1.76	21.96	22.18	2.40	0.06
SO03	200	3.27	54.38	0.02	0.56	12.21	10.15	29.83	1.95	25.28	27.20	2.65	0.08
SO04	188	0.00	23.45	0.15	0.05	48.88	42.63	15.66	0.15	10.52	24.23	1.28	0.00
SO05	217	0.00	58.88	0.02	0.17	11.11	19.23	27.43	0.09	31.06	14.93	3.00	0.00
SO06	233	0.00	61.20	0.32	0.20	15.22	22.75	29.35	0.15	34.50	17.26	3.64	0.00
SO07	225	0.00	78.84	3.84	0.09	73.15	62.21	41.25	6.13	44.85	25.77	5.29	ND
SO08	182	0.00	56.26	0.00	0.32	13.61	14.11	31.83	1.65	32.22	28.19	2.52	ND
SO09	162	0.00	19.65	0.09	0.03	42.32	39.28	13.65	0.11	10.10	21.86	1.03	0.01
SO10	174	0.00	23.03	0.11	0.04	49.77	43.38	15.54	0.14	10.91	23.26	1.10	0.01
SO11	196	0.00	27.61	0.17	0.06	53.12	48.90	18.28	0.17	12.72	27.30	1.45	0.03
SO12	236	0.00	92.01	3.97	0.012	87.68	68.10	45.00	6.41	48.91	27.63	7.61	ND
SO13	142	1.15	71.30	0.31	0.07	23.27	51.15	33.74	2.27	18.15	28.10	44.75	ND
SO14	127	1.07	68.73	0.27	0.05	20.86	49.25	12.38	2.12	16.94	26.38	42.16	ND
SO15	131	2.00	79.74	0.43	0.06	23.83	54.00	14.61	2.96	19.27	29.39	44.00	ND
SO16	152	2.06	81.36	0.61	0.007	27.15	59.24	16.71	3.04	22.62	31.29	47.27	ND
SO17	167	2.73	77.35	0.011	0.26	8.65	6.99	24.20	1.39	18.81	20.85	2.65	0.03
SO18	251	0.00	98.26	4.65	0.026	96.32	72.62	49.12	7.12	51.25	29.17	8.15	ND
SO19	239	0.00	91.95	4.34	0.019	92.70	75.15	42.74	0.32	18.70	32.70	4.65	ND
SO20	211	0.00	21.93	0.17	0.19	15.73	37.86	11.96	0.18	13.30	27.33	3.96	0.00
SO21	243	0.00	26.25	0.21	0.25	17.28	41.26	14.10	0.21	15.11	29.54	4.18	0.00
SO22	142	0.00	56.58	0.00	0.19	8.78	9.73	28.62	0.75	32.96	15.21	2.03	0.00
SO23	153	0.00	63.61	0.00	0.11	9.26	10.16	30.22	0.94	36.17	16.56	2.10	ND
SO24	165	0.00	71.20	0.00	0.93	11.83	13.27	35.45	1.26	41.12	19.27	2.65	ND
SO25	162	0.00	69.73	0.00	0.84	11.24	12.69	32.53	1.09	39.26	16.92	2.35	ND
SO26	185	2.85	55.16	0.025	0.58	12.45	9.83	25.85	1.75	26.25	26.66	2.70	0.09
SO27	198	3.17	53.69	0.045	0.52	11.26	10.65	28.22	1.83	23.91	24.15	2.56	0.06
SO28	165	0.00	49.23	0.00	0.23	9.85	10.54	25.64	0.87	28.37	19.73	1.87	ND
SO29	171	0.00	52.93	0.00	0.25	10.31	12.11	28.47	1.02	30.18	21.39	2.06	ND
SO30	193	3.12	52.94	0.019	0.49	11.83	10.11	27.30	2.04	24.77	25.32	2.50	0.07



Sulphate ions (SO_4^{2-}) values vary from 14.93 mg/L in well SO05 to 32.70 mg/L in well SO19 with average value of 24.86 mg/L; which are less than the WHO (2021) recommended limit. High concentration of SO_4^{2-} may increase the levels of hardness in groundwater repositories, may corrode plumbing done with copper and can lead to irritation of the lungs which cause serious disease of the lung (Sharma, et. al., 2020). Having low SO_4^{2-} concentrations, the studied groundwater do not pose these hazards. CO_2 values vary between 1.07 mg/L in well SO14 to 3.27 mg/L in well SO03 with a mean of 2.50 mg/L. Wells SO05, SO06, SO07, SO08, SO09, SO10, SO11, SO12, SO19, SO20, SO21, SO22, SO23, SO24, SO25, SO28 and SO29 recorded zero values. The amount of dissolved CO_2 in groundwater influences the reaction of magnesium in solution (Saha et al. 2019). The measured total hardness in the sampled wells varies from 19.65 mg/L in well SO09 to 98.26 mg/L in well SO18 with a mean of 57.80 mg/L. Total hardness reflects the relationship between cation and bicarbonate in water and ion content. All the water samples are within the standard limits. Generally, it was observed that all the sampled groundwater was below threshold recommended for Cl^- (250 mg/L), SO_4^{2-} (250 mg/L) and HCO_3^- (500 mg/L); while 24% of the wells have a concentration of NO_3^- that exceed the (50 mg/L) recommended limit in drinking water (WHO, 2021).

The cations analyzed include; calcium ions (Ca^{2+}), magnesium ion (Mg^{2+}), ferric ions (Fe^{2+}), fluoride ions (F^-), sodium ions (Na^+), potassium ions (K^+) and copper ions (Cu^{2+}). The concentration of calcium ions (Ca^{2+}) in the water samples ranges from 10.10 mg/L in well SO09 to 51.25 mg/L in well SO18; with an average of 23.90 mg/L; all the Ca^{2+} values do not exceed the WHO (2021) standard for drinking water. Presence of Ca^{2+} may be mainly due to geochemical processes involving Calcium dissolution in rocks. Calcium is a significant determinant of water hardness due its buffering qualities and functions as a pH stabilizer (Singh, et al., 2020).

Magnesium ion (Mg^{2+}) concentration varies from 11.96 mg/L in well SO20 to 49.12 mg/L in well SO19 with an average of 26.76 mg/L; which are within the WHO (2021) standard for drinking water. Both calcium and magnesium are found in groundwater that has come in contact with certain rocks and minerals. When these materials are dissolved, they release calcium and magnesium. The concentration of Mg^{2+} in groundwater system in the study area can be attributed to the dissolution of Mg ions in the groundwater system (El-Osta, et. al., M; 2022). The concentration ferric ions (Fe^{2+}) in the water samples ranges from 0.011 mg/L in well SO17 to 4.65 mg/L in well SO18 with an average value of 0.86 mg/L.

Iron concentration in water samples from wells SO06, SO07, SO12, SO13, SO15, SO16, SO18 and SO19 exceeded the WHO (2021) recommended limits. This is due volcanic rocks that characterized the underlying geology of the study area (Alghamdi, et. al., 2020). Wells SO08, SO22, SO23, SO24, SO25, SO28 and SO29 did not record ferric ion content at all. High concentration of Fe^{2+} may be as a result of the ferrogenous rocks, which contain the minerals haematite, limonite and goethite in the groundwater repositories (Akakuru, et al. 2022; Ibuot, et al. 2017; Urom, et al. 2021). Fluoride ion (F^-) concentration varies from 0.007 mg/L in well SO16 to 0.93 mg/L in well SO24 with an average of 0.23 mg/L; which are low and below the recommended W.H.O (2021) limit. Sodium ion (Na^+) concentration ranges from 0.09 mg/L in well SO05 to 7.12 mg/L in well SO18 with an average value of 1.71 mg/L; which are within the WHO (2017) permissible limits. Potassium ion (K^+) concentration varies from 1.03 mg/L in well SO09 to 47.27 mg/L in well SO16 with an average of 8.58 mg/L; which are within the WHO (2021) permissible limits. Potassium and sodium are monovalent ions and as such do not cause



hardness of water. The absence of potassium will affect the normal functioning of the body cells (Urom, et al. 2021). Copper ion (Cu^{2+}) ranges from 0.01 mg/L in wells SO09 and SO10 to 0.09 mg/L in well SO26 with an average of 0.049 mg/L; which are far less than the WHO (2021) permissible limit. Wells SO04, SO05, SO06, SO20, SO21 and SO22 recorded no Copper ions and wells SO06, SO07, SO12, SO13, SO14, SO15, SO16, SO18, SO19, SO23, SO24, SO28 and SO29 were not determined.

From the results, it can be observed the major anions chlorides, bicarbonates, nitrite and sulphates while the major cations are calcium, magnesium and potassium. Sodium, (Na^+) Iron (Fe^{2+}), Fluoride (F^-) and Copper (Cu^{2+}) ions seem not to be too high. However, analysis of the cations show that $\text{Mg}^{2+} > \text{Ca}^{2+} > \text{K}^+ > \text{Na}^+$ decrease in that order while anions analysis shows $\text{HCO}_3^- > \text{NO}_3^- > \text{Cl}^- > \text{SO}_4^{2-}$ decrease in that order. The Hydro-geochemistry of the cations which reveals that Ca^{2+} , Mg^{2+} , HCO_3^- , NO_3^- , Cl^- and SO_4^{2-} possessing the highest regional distribution and concentration in the study area is an indication of hardness due to the presence of the magnesium bicarbonate, magnesium sulphate and calcium chloride.

Heavy Metals

The heavy metals analysis in the thirty (30) wells in the field showed that heavy metals values fall within the accepted WHO Standard values. Analysis showed that most of the analyzed water samples had little or no heavy metal content except for iron which have slight values (Table 2). These observed values are not due to the presence of pollution within the groundwater but rather due to the presence of ferromagnesian minerals in the rocks that dominate the study area (Yahaya, et. al., 2021).

Table 3: Groundwater quality in the study area compared to World Health Organization WHO Standard

Parameters	Minimum	Maximum	Average	WHO (2021) Standard
Temperature ($^{\circ}\text{C}$)	18.24	21.52	19.71	30 – 35
pH	4.6	6.3	5.43	6.5 – 8.5
Electrical Conductivity ($\mu\text{S}/\text{cm}$)	142	452	249.95	500
Total Dissolved Solids	110	301	153.63	500
Sodium	0.09	7.12	1.71	200
Total Hardness (mg/L)	19.65	98.26	56.80	500
Potassium (mg/L)	1.03	47.27	8.58	12
Magnesium (mg/L)	11.96	49.12	26.76	50
Calcium (mg/L)	10.10	51.25	23.90	75
Bicarbonate (mg/L)	142	251	185.87	500
Chlorite (mg/L)	8.65	96.32	28.01	250
Nitrate (mg/L)	6.99	75.15	30.47	50
Sulphate (mg/L)	14.93	32.70	24.86	250
Fluorite (mg/L)	0.007	0.93	0.23	1.5
Iron (mg/L)	0.011	4.65	0.86	0.3
Copper (mg/L)	0.01	0.09	0.049	250



Suitability for Irrigation Purpose

The suitability of groundwater in the study area for irrigation purposes was examined and evaluated according to World Health Organization (WHO, 2021) standards. When using groundwater for irrigation, the accumulation of salts and alkalinity in the soil must be taken into consideration. Electrical conductivity (EC), total dissolved solids (TDS), total hardness (TH), sodium adsorption ratio (SAR), magnesium hazard (MH), Kelly's ratio (KR), percentage sodium (%Na⁺) and permeability index (PI) are the irrigation water quality indices used to examine the possible risks.

Sodium Adsorption Ratio (SAR)

Sodium Adsorption Ratio (SAR) defines the agricultural water quality status and is used to study soil sodium hazards. It is an essential parameter for irrigation water quality assessment which involves the relationship between the ions of sodium versus calcium and magnesium (Paternoster et al. 2021). The results in Table 1 were used to estimate the SAR as presented in Table 3. High SAR is an indication of low water quality for irrigated agriculture use. This study reveals that SAR concentration varies from 0.41–2.78 with a mean of 1.47. According to Paternoster et al. 2021), SAR values that are less than 10 (< 10) indicate that all the water samples have excellent water quality for irrigation.

Magnesium Hazard (MH)

Magnesium is necessary for the formation of chlorophyll molecules in plant tissue and also in activation of specific enzymes. Its availability in groundwater depends on aquifer minerals, weathering and climate (Urom, et al. 2021). Magnesium Hazard (MH) is determined using the Ca²⁺ and Mg²⁺ average concentrations. Prediction of MH parameter in irrigation water highlights its suitability for irrigation. In this study, the MH values ranged from 42.22 in well SO13 to 69.56 in well SO19 with a mean 51.19. Khodapanah et al. (2009) opined that MH > 50 is not recommended for irrigation. Hence, water samples from wells SO01, SO02, SO03, SO04, SO09, SO10, SO11, SO13, SO17, SO19, SO27 and SO30; having values greater than 50 are consequently are not recommended for irrigation purposes.

Kelly's Ratio (KR)

Kelly's Ratio is obtained from the values of sodium against the sum of calcium and magnesium ions. The results in Table 3 show the values of KR vary from 0.0015 in well SO05 to 0.0874 in well SO15 with an average of 0.0300. KR > 1 indicates an excess level of Na⁺ in waters. Water having KR value above one is considered unsuitable water for irrigation since it implies a significant quantity of sodium ions, while the one with values below one is suitable for irrigation as it specifies sodium ions deficiency (Akakuru, et al. 2022). Water samples from all the wells have values less than 1 (<1), thus are rated as suitable for irrigation purpose due to no alkali hazard.

Percentage Sodium (%Na⁺)

Sodium affects the soil by closing up the pores and this leads to poor water infiltration and wetting of the soil and can cause permeability problems (Singh et al. 2020 and Paternoster et al. 2021). The %Na index is important in determining groundwater suitability for irrigation



purpose. When %Na is < 20%, it is excellent, 20 – 40% is good, 40 – 60% is permissible, 60 – 80% is doubtful and > 80% is unsuitable (Akakuru, et al. 2022). The results in Table 3 show %Na values ranged from 0.15% in well SO05 to 6.29% in well SO07 with a mean value of 14.09%. This implies that water samples from all the wells are within the permissible category of excellent (<20%) for irrigation (Akakuru, et al. 2022).

Permeability Index (PI)

Permeability index (PI) is calculated based on the values of Ca^{2+} , Mg^{2+} , Na^+ and HCO_3^- dissolution in the groundwater system (Omorogieva, et. al., 2022). From Table 3, values vary from 14.9 in well SO16 to 57.8 in well SO20 with an average of 31.2. Values less than 25 (<25.0) are unsuitable, 25.0–75.0 are good, while values exceeding 75(>75.0) are suitable (>75.0) for irrigation (Li et al., 2021). Hence, groundwater from wells SO06, SO07, SO08, SO12, SO16, SO18, SO22, SO23, SO24, SO25 and SO29 are unsuitable. The remaining, making about 64% of the studied groundwater are good for use in irrigation.

Electrical Conductivity (EC), Total Dissolved Solids and Total Hardness

In this study, the Electrical Conductivity (EC) and Total Dissolved Solids (TDS) results show an appreciable level of acceptance for drinking purposes but continued dissolution of materials in the groundwater system can cause unusual level of EC and TDS respectively (Akakuru, et al. 2022). EC indicates the total salt concentration, which is dependent on the charges from dissolved ions in water. It is a measure of salinity hazard to crops; excess salinity reduces the osmotic activity of plant and thus interferes with the absorption of water and nutrients from the soil (Singh et al. 2020). A high EC of water indicates a high concentration of ions, representing the higher TDS in water. The EC values of the studied water samples are low and fall within the recommended standards for irrigation practice. TDS quantifies the materials such as sulfate, bicarbonate, sodium, carbonate, calcium, magnesium dissolved in water. When the TDS increase, it leads to reduces water clarity, giving the water a color and salinity behavior (Paternoster et al. 2021). The Total Hardness (TH) is an indicator of salt accumulation in the soil. Plant growth is directly affected by the salt content of irrigation water, and indirectly by the permeability of the soil structure and aeration.

Water pH and temperature

Water pH influences the volatility of the processes in flocculation and coagulation, hence affecting emitter clogging (Akakuru, et. al., 2022). At a pH higher than 7.2, along with a high temperature, the magnesium and calcium carbonates precipitate in the lateral tubes and the drippers since the carbonates' solubility is inversely proportional to temperature. Most surface drip irrigation systems have tubes painted black and exposed to the surface; hence, they can be heated by the sun to high temperatures (Akakuru, et. al., 2022). Consequently, a combination of heat from the sun and a pH higher than 7.2 precipitates calcium and magnesium carbonate in water; accumulating within the emitters. The groundwater under study recorded maximum temperature and pH of 21.52 °C and 6.3 respectively. Hence the studied groundwater water is free from risks related to flocculation and coagulation resulting in clogging of irrigation system.



Conclusions

Groundwater in Song and environs is moderately acidic and unsuitable for direct drinking without treatment in some wells. Most physicochemical parameters meet WHO standards, but elevated nitrate and iron pose health concerns. For irrigation, most samples are suitable based on SAR, KR, and %Na⁺, but 40% are unsuitable due to high MH. Regular monitoring is recommended to prevent deterioration from anthropogenic activities.

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